

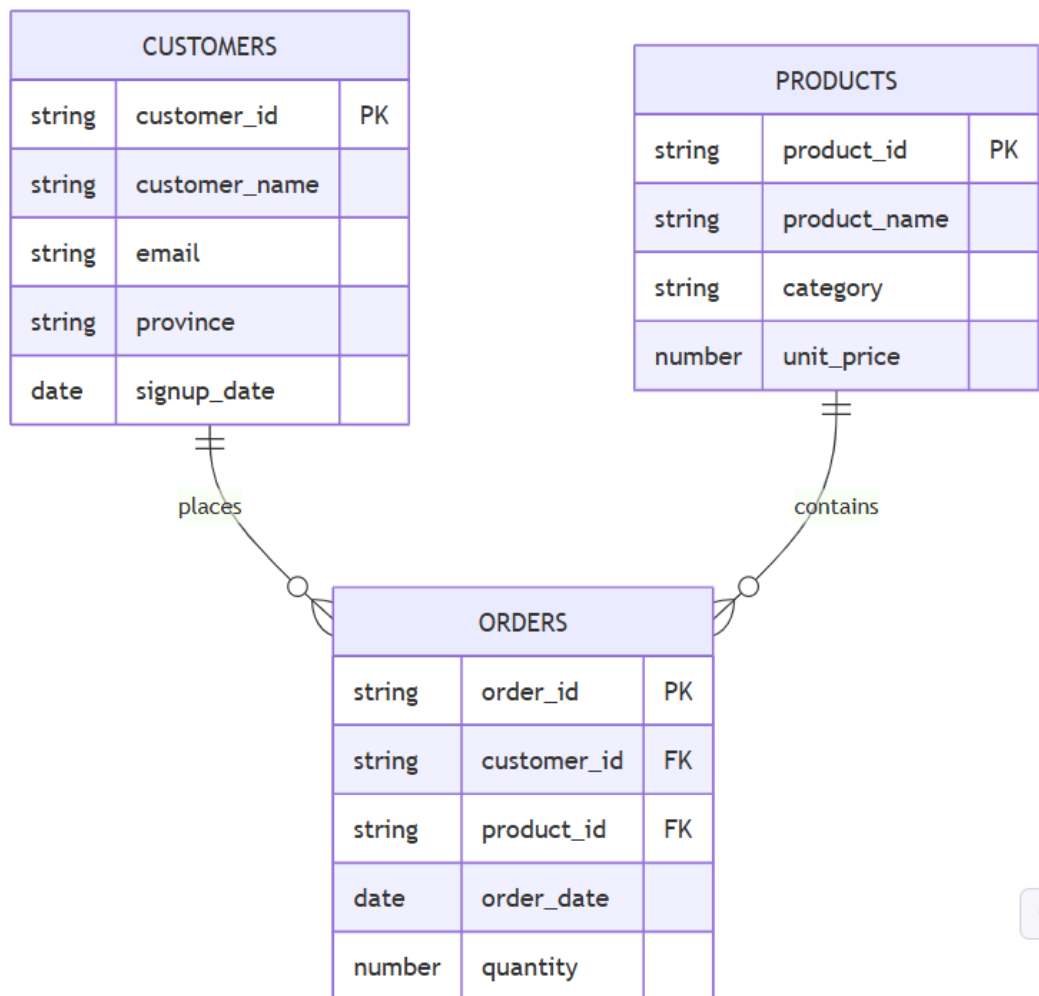
Data Engineering Project 1

Platform: Snowflake only. Work done in any other tool will not be marked.

Data provided:

File	Rows	Columns	Role
customers.csv	50	customer_id, customer_name, email, province, signup_date	Dimension
products.csv	20	product_id, product_name, category, unit_price	Dimension
orders.csv	150	order_id, customer_id, product_id, order_date, quantity	Fact

Data model



What you need to do

1. Create a database and schema in Snowflake for this project. Name it clearly, for example DE_PROJECT1.
2. Load all three CSVs into Snowflake tables. Use correct data types per column, not VARCHAR for everything.
3. Confirm your load. Run `SELECT COUNT(*)` on each table and check: customers = 50, products = 20, orders = 150.
4. Write and run these four queries, and keep the SQL for each:
 - Every order joined to the customer name, product name, category, and a calculated line_revenue (quantity times unit_price).
 - Total revenue per customer.
 - Total revenue per product category.
 - Top 5 customers by total spend.
5. Write one short paragraph per query explaining what it shows and why it matters.

What gets marked

Item	Points
Database and tables created in Snowflake, correct types	20
Successful load, row counts confirmed	15
Query 1: order detail join	15
Query 2: revenue per customer	15
Query 3: revenue per category	15
Query 4: top 5 customers	10
Write-up quality (clear, correct, ties back to the data)	10
Total	100

What to submit

- Screenshot of your Snowflake database, schema, and tables

- The load statements you used (COPY INTO or equivalent)
- A .sql file with all four queries
- Results for each query (screenshot or CSV export)
- Your write-up paragraphs

Deadline: 14 September 2026. Late work without prior arrangement is not marked.

This is Project 1. It feeds directly into the capstone (BrightLearn_Snowflake_Capstone.md), which assumes you can already load and join data in Snowflake on your own.